INCH-POUND

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5 May 1995
SUPERSEDING
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(See 6.12)

MILITARY SPECIFICATION

HEATER SYSTEMS, WATER, STEAM HEATED NON-STORAGE TYPE

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification describes quick recovery, steam water heaters as a complete system. This specification ensures that all system components are compatible and properly integrated with regard to size, function, and performance. The heater systems comply with system integration requirements, performance, endurance, safety test procedures, shock and vibration, and standardization for installation on U.S. Naval Surface ships.

1.2 Classification. Heaters are of the following classes, as specified (see 6.2):

Classes:

- Class A - Personnel use, 125 degrees Fahrenheit (°F) (example: showers and lavatories).
- Class B - Non-personnel use, 160°F (example: grease intercept hoods and pot sanitizing sinks).
- Class C - Non-personnel use, 180°F (example: laundry and scullery).

1.2.1 Temperature rating. The temperature rating of heater systems provided in accordance with this specification shall be 125°F, 160°F, and 180°F for Class A, B and C respectively. These ratings are synonymous with the class designation throughout.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, SEA 03R42, Naval Sea Systems Command, 2531 Jefferson Davis Hwy, Arlington, VA 22242-5160 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1.2.2 Capacity. The capacity of heater systems provided in accordance with this specification shall be rated based on the amount of outlet water that can be generated at rated temperature assuming a 40°F inlet water temperature. Capacity ratings shall be in gallons per minute (gal/min). Ratings shall be in multiples of 5 gal/min.

1.3 Terms. The following describes terms used in this specification.

a. "Heater" and "heater system" are used synonymously within this specification to mean the complete assembly of heat exchanger, control, monitoring, and miscellaneous equipment supplied.

b. "Load" refers to the demand for flow of outlet water. Full load condition is when heater is required to deliver the rated capacity.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

WW-S-2739	-	Strainers, Sediment: Pipeline, Water, Air, Gas, Oil, or Steam.
WW-T-696	-	Traps, Steam.
PPP-F-320	-	Fiberboard; Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes.

MILITARY

MIL-S-901	-	Shock Tests, H.I. (High-Impact) Shipboard Machinery, Equipment and Systems, Requirements for.
MIL-E-917	-	Electric Power Equipment, Basic Requirements (Naval Shipboard Use).
MIL-F-1183	-	Fittings, Pipe, Cast Bronze, Silver-Brazing, General Specification for.
MIL-S-1222	-	Studs, Bolts, Hex Cap Screws, Socket Head Cap Screws and Nuts.
MIL-S-4040	-	Solenoid, Electrical, General Specification for.
MIL-P-15024	-	Plates, Tags, and Bands for Identification of Equipment.
MIL-P-15024/5	-	Plates, Identification.
MIL-I-17244	-	Indicators, Temperature, Direct-Reading, Bimetallic, (3 and 5 Inch Dial).
MIL-E-17555	-	Electronic and Electrical Equipment, Accessories, and Provisioned Items (Repair Parts): Packaging of.

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MILITARY (Continued)

MIL-G-18997	-	Gauge, Pressure, Dial Indicating.
MIL-L-19140	-	Lumber and Plywood, Fire-Retardant Treated.
MIL-V-19772	-	Valves, Temperature Regulating, Automatic, Naval Shipboard.
MIL-F-20042	-	Flanges, Pipe and Bulkhead, Bronze (Silver Brazing).
MIL-T-24270	-	Thermowells for Thermometers and Electrical Temperature Sensors General Specification for.
MIL-G-24696	-	Gasket, Sheet, Non-Asbestos.

STANDARDS

MILITARY

MIL-STD-167-1	-	Mechanical Vibrations of Shipboard Equipment (Type I - Environmental and Type II - Internally Excited).
MIL-STD-278	-	Welding and Casting Standard.
MIL-STD-454	-	Standard General Requirements for Electronic Equipment.
MIL-STD-461	-	Requirements for the Control of Electromagnetic Interference Emissions and Susceptibility.
MIL-STD-769	-	Thermal Insulation Requirements for Machinery and Piping.
MIL-STD-889	-	Dissimilar Metals.
MIL-STD-1399, Section 300	-	Interface Standard for Shipboard Systems Section 300 Electric Power, Alternating Current. (Metric)
MIL-STD-1686	-	Electrostatic Discharge Control Program for Protection of Electrical and Electronic Parts, Assemblies and Equipment (Excluding Electrically Initiated Explosive Devices). (Metric)
DOD-STD-2169	-	High Altitude Electromagnetic Pulse Environment.
MIL-STD-45662	-	Calibration System Requirements.

HANDBOOKS

MILITARY

MIL-HDBK-263	-	Electrostatic Discharge Control Handbook for Protection of Electrical and Electronic Parts, Assemblies and Equipment (Excluding Electrically Initiated Explosive Devices). (Metric)

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, BLDG. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

PUBLICATIONS

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NAVAL SEA SYSTEMS COMMAND (NAVSEA)
0900-LP-001-7000 - Fabrication and Inspection of Brazed Piping Systems.

(Application for copies should be addressed to the Standardization Documents Order Desk, BLDG. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)
Z 21.22 - Relief Valves and Automatic Gas Shutoff Devices for Hot Water Supply Systems. (DoD adopted).

(Application for copies should be addressed to the American National Standards Institute, 11 West 42nd Street, 13th Floor, New York, NY 10036.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)
B 16.5 - Pipe Flanges and Flanged Fittings.
Boiler and Pressure Vessel Code, Section IV - Rules for Construction of Heating Boilers.

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)
A 36 - Standard Specification for Structural Steel. (DoD adopted)
A 216 - Standard Specification for Steel Castings, Carbon, Suitable for Fusion Welding, for High-Temperature Service. (DoD adopted)
A 516 - Standard Specification for Pressure Vessel Plates, Carbon Steel, for Moderate- and Lower-Temperature Service. (DoD adopted)

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.5) as specified in 4.3.

3.2 Materials. The materials that shall be used in manufacturing the heaters are listed in table I. This table is not restrictive and materials of equal or better service may be proposed, providing approval has been granted for the use of any alternate materials (see 6.8). Material compatibility shall be in accordance with MIL-STD-889.

TABLE I. Steam water heater system material list.

Part	Material	Specification 1/
Water side of system (parts in contact with water) except coils, coil manifolds, and non-metallic elements (see below)	Corrosion resistant steel Copper-nickel alloy Copper Bronze	 UNS C70600 UNS C71500
All parts on steam side of heater (in contact with steam or condensate except coils and coil manifolds (see below)	Corrosion resistant steel Carbon steel Copper-nickel alloy Copper Bronze	 ASTM A 36 ASTM A 216 ASTM A 516 GR 55 UNS C70600 UNS C71500
Gasket, sheet	Flexible graphite	MIL-G-24696
O-rings and other non-metallic elements	Viton Approved elastomers	
Bolting	Alloy steel Stainless steel nickel-copper alloy	MIL-S-1222 grades B7, 2H, 304, 316, 400
Coils and coil manifolds	Corrosion resistant steel Copper-nickel alloy Copper Bronze	 UNS C70600 UNS C71500

1/ Specifications for materials shall be in accordance with ASME BPVC Section IV. Specifications are noted herein only where ASME BPVC Section IV is silent or less stringent.

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3.2.1 Corrosion protection.

3.2.1.1 Corrosion-resisting materials. Corrosion-resisting steel, copper, bronze, copper-nickel, and copper-nickel alloys as specified herein are considered corrosion-resisting materials. Where corrosion-resistant steel is used, it shall be 300 series stainless steel. Corrosion-resisting steel, when fabricated by any method that tends to reduce corrosion-resistant properties, shall be normalized to restore those properties before being used in the assembly of any unit.

3.2.1.2 Non-corrosion-resisting materials. Parts other than corrosion-resisting materials shall be protected against corrosion by the use of chemicals, electrolytic processes, plating, or paints and enamels.

3.2.1.3 Hardware. Self-tapping screws with machine-screw threads may be used in the cabinet assembly. Sheet-metal screws with sheet-metal threads shall not be permitted.

3.2.1.4 Recovered materials. Unless otherwise specified herein, all equipment, material, and articles incorporated in the products covered by this specification shall be new and may be fabricated using materials produced from recovered materials to the maximum extent practicable without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products is allowed under this specification unless otherwise specifically specified.

3.3 Construction.

3.3.1 General requirements. The steam water heater system shall be a complete system for field assembly or pre-assembly in which all components and materials are compatible and properly integrated with regard to size, function, and performance (see 3.9 and 6.3). The heaters shall be the quick recovery type utilizing steam as the heat source on either the shell or tube side. In addition, the heater shall heat or discharge water at the recommended temperature and capacity upon demand. Heater systems shall have good accessibility to all internal and external parts for ease of serviceability and cleaning. This requirement shall prohibit the location of inlet and outlet lines and auxiliary components from obstructing flange end connections and other field removable parts. For example, the heat exchanger steam and water connections shall be accessible when installed on shipboard to permit both ease of inspection and removal, cleaning, or repair of the following: coil assemblies, cover plates, shell, shell liners, couplings, and so forth. Auxiliary water heater system components shall be accessible to permit ease of inspection, tool maneuverability, and removal and/or repair of these components. See figure 1 for a typical Class A steam water heater system. See figure 2 for a typical Class B or C steam water heater system. The system shall include all components defined in system envelopes on figures 1 and 2. The equipment shall reflect good serviceability as specified in 3.6.13.5.

3.4 Reliability requirements. Each heater system shall have an operating life of 100,000 hours. The water heater systems shall have a mean time between failures (MTBF) of 10,000 hours of operation, mean time to repair (MTTR) of 12 hours (see 4.6.8), mean time between corrective maintenance (MTBCM) of 1,000 hours, and an

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operational availability (Ao) of 0.92 (see 6.3). The MTBF shall be derived by analytical or empirical means.

3.5 Temperature adjustments. Class A, class B, and class C heaters shall have an adjustable outlet water temperature range of 100 to 125°F, 120 to 160°F, and 140 to 180°F respectively. The outlet temperature shall be maintained automatically to within plus or minus 5°F of any setting within the adjustable temperature range, and any operating conditions within the limits specified in 3.7.1.

3.5.1 Anti-tampering device. When specified (see 6.2) for class A heaters, the contractor shall provide an approved (see 6.8) anti-tampering device to prevent unauthorized personnel from adjusting the temperature controls.

3.6 Fabrication. The water heater system, including the steam and water sides, and connections thereto, shall be constructed for a maximum design pressure of not less than 150 pounds per square inch (lb/in^2) gauge.

3.6.1 Welding and silver brazing. Welding shall conform to MIL-STD-278. In addition, qualification of welders and brazers shall be in accordance with MIL-STD-278 and NAVSEA 0900-LP-001-7000, respectively.

3.6.2 Heat exchanger. The heat exchanger shall consist of a heating chamber, heat transfer element, inlet and outlet piping connections, and miscellaneous connections as specified herein. The heating chamber shall consist of a shell with heads or cover plates. Welded fabrication of heat exchanger components shall meet be in accordance with MIL-STD-278, class A-4. The heat exchanger shall be furnished with a base support that is either welded or bolted to the heating chamber. The base support shall be deck mounted by a minimum of four bolts with 5/8 - 11UNC thread.

3.6.2.1 Shell. Shell shall be of seamless castings, seam welded rolled sheet, or forged.

3.6.2.2 Head or cover plates. Head or cover plates shall be either cast or forged and shall be attached to shell by bolted flange. The head or cover plate material shall be as specified in table I.

3.6.2.3 Shell liners. If used, shell liners shall be fabricated from sheet material either silver-brazed or seal welded. Liners shall be a thickness of not less than 1/32 inch. Liners and flanges for liners shall be assembled to prevent interior surfaces of the shell from being exposed to liquid. Liners shall fit securely inside the shell without wrinkling or buckling. Cladding of the shell will not be accepted.

3.6.2.4 Heat transfer element. The heat transfer element shall be either coils, U-tube bundle, or straight bundle utilizing steam as a heat source on either the shell or tube side. If coils are furnished, they shall be supported securely within the structure of the heat exchanger. The coils shall be placed (with framework or brackets), so that when steam is circulated inside the coils, full drainage will occur when the heater is in the upright position. Air traps shall not interfere with circulation. The coils shall be mounted in either a pancake or spiral nest. If a U-tube heating element or straight bundle is furnished, the element, U-tube bundle, or straight bundle shall be securely supported in a partial shell or wrapper extending

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into the heating chamber. The U-tube bundle or straight bundle shall be arranged to permit full drainage of the condensate. If steam is utilized on the shell side, the condensate drain connections shall be arranged to permit full drainage of the condensate.

3.6.3 Inlet steam pressure. Heat exchanger shall be designed for a steam inlet pressure of 40 ± 10 lb/in^2 gauge. When specified for a use with a steam system exceeding 50 lb/in^2 gauge nominal, the heat exchanger shall be designed for a steam pressure of (nominal pressure minus 10) ± 10 lb/in^2 (see 3.7.1(f) and 6.2).

3.6.4 Surface temperature. Heater components and piping whose surface temperature under normal operation can exceed 125°F shall be insulated in accordance with MIL-STD-769. All mechanical joints shall be provided with removable insulation.

3.6.5 End connections. Unless otherwise specified (see 6.2), end connections for all steam and water inlets and outlets shall terminate in take down connections. The water inlet, outlet, and recirculation connections (see 3.6.12) shall be female silver-brazed union end connections with o-ring and retaining ring seal in accordance with MIL-F-1183, type A. Depending on whether the pipes leading to and from the heat exchanger are steel or non-ferrous material, the steam supply and condensate drain connections shall be flanged, either with steel raised or flat faced flanges in accordance with ASME B16.5, class 150 or bronze flanges in accordance with MIL-F-20042, class 150. Threaded pipe connections shall not be used. Heat exchanger shall be equipped with necessary drain valve or valves to permit complete drainage of the heating chamber.

3.6.6 Relief valve boss. Heat exchangers utilizing steam on the tube side and water on the shell side shall be equipped with boss for the relief valve. Heat exchanger shall be equipped with boss to accommodate the relief valve's ground joint union connection with tailpiece.

3.6.7 Thermal bulb connection. Heater systems that utilize thermal bulbs (part of steam regulating valve) shall have a 1-1/8 - 12UN straight threaded connection to accept the thermal bulb retaining nut. Mating face shall be ground flat to accommodate either an O-ring or washer, to form a tight seal.

3.6.7.1 Thermal bulb boss. Heat exchangers that utilize steam on the tube side and water on the shell side shall be equipped with a boss for the steam control valve's thermal bulb. This boss shall be integral with the heating chamber (shell or head).

3.6.7.2 Thermal bulb well or shroud. Heat exchangers that utilize steam on the shell side and water on the tube side shall be equipped with a well or shroud for the steam control valve's thermal bulb. The well or shroud shall be integral with the heat exchanger and shall allow sufficient water flow for thermal bulb to sense the true temperature of the heated water and activate. The well or shroud shall provide positive alignment and allow easy insertion of the thermal bulb into the well from either a vertical or horizontal position. Piping connected to the well shall be supported on both sides of the well.

3.6.8 Heater supports. The heater system shall include a frame to support and elevate all heater components to a height permitting all connections to be easily

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accessible without having to remove framework or any other obstruction (see 3.6.1). Contractor shall identify any clearance requirements necessary for accessibility.

3.6.9 Identification and information plates. Heaters shall have an identification plate mounted on the shell of the heater in full view and it shall contain the following:

(a) Manufacturer's name and address.
(b) Manufacturer's model, serial, and part numbers.
(c) Steam pressure rating (lb/in^2 gauge).
(d) Class and rated capacity (gal/min) (see 1.2).
(e) Contract or purchase order number.
(f) National stock number.
(g) Rated temperature (°F) (see 1.2.1).

Identification plates shall be in accordance with type C of MIL-P-15024 and MIL-P-15024/5.

3.6.10 Controls, safety devices, and indicators. Controls, safety devices, and indicators, as shown on figures 1 and 2 shall be furnished as part of the water heater system.
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3.6.10.1 Steam regulating valve. Steam regulating valves shall be self-contained in accordance with MIL-V-19772 and shall not rely on compressed air or other external power media for controlling the flow of steam to the heat exchanger. The regulating valve may regulate on the basis of temperature or delivered water flow demand and shall be as specified in the design, material, and construction requirements of MIL-V-19772 for type II, class 1 and type III, class 1 valves. Also, the regulating valve shall have the capability to completely secure steam to the heat exchanger when signaled by the overtemperature safety control system (see 3.6.10.2 and 3.6.10.7).

3.6.10.2 Overtemperature safety shutdowns. Class A heater systems shall have two independent features utilized to prevent overtemperature situations from occurring in the delivery water systems. One secures the flow of steam, and the other secures the flow of outlet water. Class B and C heaters do not require an overtemperature safety shutdown system. Solenoids shall be in accordance with MIL-S-4040 (see 3.6.11).

3.6.10.2.1 Safety shutdown (Steam side). A solenoid shutoff valve shall be installed in the pilot supply to the steam regulating valve (see figure 1). The valve shall be electrically interconnected to the overtemperature safety control system (see 3.6.10.7), have an independent temperature sensor located in the outlet water line downstream of the recirculation connection, and employ an automatic reset to avoid frequent trips and to control the temperature of the outlet water. The solenoid valve when actuated to close (de-energize) shall shut off steam supply to the pilot and thereby cause the steam regulating valve to shut down tightly.

3.6.10.2.2 Safety shutdown (Water side). A solenoid shutoff valve shall be installed in the outlet water line from the heat exchanger (see figure 1). The valve shall be electrically interconnected to the overtemperature safety control system and have an independent temperature sensor in the outlet water line, downstream of the steam safety shutdown temperature sensor (see 3.6.10.2.1), and employ an automatic

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reset to secure the delivered water system on overtemperature conditions. The solenoid valve when actuated to close shall shut off the outlet water supply to the delivery system. A normally closed contact shall be supplied to allow a recirculation system to sense valve actuation. This contact shall open when valve is actuated to close, and shall be rated at 115 VAC.

3.6.10.3 Pressure relief valve. Heat exchangers utilizing water on the shell side shall have an integral pressure relief valve for thermal expansion or system over pressurization in accordance with ANSI Z21.22 and the ASME BPVC Section IV. The relief valve shall be provided with a ground joint union connection, with a tailpiece for brazing. The relief valve shall be set to relieve at 150 ± 3 lb/in^2 gauge.

3.6.10.4 Steam traps and orifices. Steam traps and orifices shall be provided as necessary to assure proper operation of the heater system as specified in 3.7.1.

3.6.10.4.1 Steam traps. Steam traps shall be in accordance with WW-T-696 and sized to ensure adequate condensate (or moisture) drainage in the heater steam supply and discharge lines. A thermodynamic disc type of steam trap, type III, class 2, with integral strainer shall be located before and after the steam regulating valve to ensure condensate is at a minimum. In addition, one float and thermostatic type of steam trap, type VI, class 2, shall be located in the condensate discharge line of the heater system and it shall be sized to operate properly at twice the rated capacity.

3.6.10.4.2 Orifices. Orifices shall be sized by the heater manufacturer for maximum heater capacity and for the proper application.

3.6.10.5 Strainers. Strainers for the protection of steam regulating valves and steam traps shall be furnished as part of the water heater system in accordance with WW-S-2739, type I or II, style Y.

3.6.10.6 Temperature and pressure gauges. Heaters shall have temperature gauges located in the steam inlet line, the water outlet line, the water inlet line, recirculation water inlet, and combined water inlet adjacent to the heat exchanger. In addition, each heater shall have pressure gauges located in the steam inlet line upstream and downstream of regulating valve, the condensate drain line, and the water inlet line.

3.6.10.6.1 Temperature gauges. Each temperature gauge shall have a direct reading 3-inch or larger dial size, in accordance with MIL-I-17244. The steam inlet temperature gauge shall have a temperature range of 50 to 750°F and all water line temperature gauges shall have a temperature range of 20 to 240°F. Thermowells shall be socket welded or brazed, 3/4-inch NPS in accordance with MIL-T-24270 and configured to M24270/1-M2CNR.

3.6.10.6.2 Pressure gauges. Each pressure gauge shall have a direct reading 4-1/2 inch or larger dial size, with range of 0 to 200 lb/in^2 gauge for water outlet line and 0 to 100 lb/in^2 for steam inlet and outlet lines in accordance with MIL-G-18997. Pressure gauges shall be mounted on a gauge panel (see 3.6.10.8).

3.6.10.7 Overtemperature safety control system. Overtemperature safety control systems are required for class A heater systems only. These units shall have an

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adjustable overtemperature safety control system in the event the outlet water temperature rises above the pre-set 125°F temperature setting. The control system shall trip at the pre-set temperature of 130 +5/-0°F and signal the steam regulating valve (via pilot solenoid valve) to secure steam flow to the heat exchanger (see 3.6.10.1 and figure 1). This pilot solenoid shutoff valve employs an automatic reset to avoid frequent trips (see 3.6.10.2.1 and figure 1). In addition, the control system shall trip out at the pre-set temperature of 135 +5/-0°F and signal the outlet water solenoid shutoff valve to secure the hot water delivery system (see 3.6.10.2.2 and figure 1). The safety device shall activate a visual alarm consisting of a red telltale light mounted on a gauge panel (see 3.6.10.8) to alert personnel of an overtemperature condition. The visual alarm shall be activated simultaneously with the trip signal to steam regulating valve. The visual alarm shall have an automatic reset similar to the control of the steam regulating valve.

3.6.10.8 Gauge panel. Pressure gauges and overtemperature safety cut-out switch and indicators shall be mounted on a gauge panel that is easily visible and accessible for repair or replacement. The panel shall be constructed for remote mounting up to at least 15 feet from the heat exchanger. The panel shall be constructed of corrosion resistant steel or treated to resist corrosion as specified in 3.2.1.2. Markings shall be in accordance with MIL-P-15024, type C. Lettering shall be 0.25 inch high and located on the gauge panel adjacent to corresponding gauge or indicator.

3.6.11 Electrical equipment. Unless otherwise specified (see 6.2), electrical equipment, such as control lights, shall operate on 120 volts (V), single phase, 60 hertz (Hz) electrical service and conform to the applicable requirements of MIL-E-917. Solenoids shall be in accordance with MIL-S-4040. Solenoids shall have continuous duty operation and shall either have a watertight housing or be encapsulated or both.

3.6.11.1 Electromagnetic emission and susceptibility. The monitoring system shall meet the requirements of MIL-STD-1399, section 300 and demonstrate compliance in accordance with requirements CE101, CE102, CS104, RE101, RE102, RS101, and RE103 of MIL-STD-461.

3.6.11.2 Electromagnetic pulse (EMP) protection. All monitoring systems produced as specified in this specification shall survive and operate after being subjected to an EMP environment as described by DOD-STD-2149.

3.6.11.3 Electrostatic discharge (ESD). All monitoring systems produced as specified in this specification will meet the ESD requirements of MIL-STD-1686, and MIL-STD-454, requirement 75, with the guidance of MIL-HDBK-263.

3.6.12 Recirculation capability. The heater system shall be capable of being directly interconnected with a hot water recirculation system. The heater system shall be furnished with recirculation connection points and all components necessary to provide a complete and safe operating system (see 3.6.10.2) when interconnected with a hot water recirculation system having a recirculating capacity and volume equal to the heater rated capacity (gal/min). The heater system shall be capable of maintaining temperature of recirculated water within rated outlet water temperature limits (see 3.7.1(h)) without the need for manual adjustment. When a three-way thermostatic valve is required for such control, a valve in accordance with MIL-V-19772, type I shall be furnished (see figure 3).

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3.6.13 Support considerations. The construction and operation of the heater system shall reflect the following support considerations to ensure the effective and economical support of the system for its life cycle.

3.6.13.1 Maintenance planning. The maintenance concept for the system shall be piece part repair by the ship's force. The system is designated as non-essential to the ship's mission (see 3.4).

3.6.13.2 Manpower and personnel. The system shall be constructed to be operated and maintained by two seamen with training equivalent to a competent journeyman with skills in plumbing and pipe fitting.

3.6.13.3 Safety. When specified (see 6.2), a preliminary hazard analysis shall be made (see 6.3).

3.6.13.4 Facilities. The system shall constructed to be repairable at any standard shipboard or shore intermediate maintenance facility capable of plumbing system repair and testing.

3.6.13.5 Serviceability. The system shall be constructed for accessibility of all components requiring servicing or replacement.

3.6.13.6 Special tools. Special tools required for servicing the system including associated valves and other major sub-components shall be furnished by the contractor.

3.7 Performance conditions.

3.7.1 Normal operating conditions. Unless otherwise specified (see 6.2), heater systems shall meet all specified performance requirements under the following parameters:

(a) Heater inclination: (see 3.7.3).
(b) Inlet water temperature: 40°F (unless otherwise specified, see 6.2).
(c) Inlet water pressure: 30 to 110 lb/in^2 gauge (unless otherwise specified, see 6.2).
(d) Load (outlet water flow condition): variable from shutoff to rated capacity (see 6.2)
(e) Inlet steam temperature: saturated (min: 287°F; max: 308°F) (unless otherwise specified, see 6.2).
(f) Inlet steam pressure: 40 ± 10 lb/in^2 gauge (unless otherwise specified, see 6.2).
(g) Condensate drain back pressure: 10 lb/in^2 gauge.
(h) Outlet water temperature: class A: 125 ± 5°F, class B: 160 ± 5°F, class C: 180 ± 5°F (unless otherwise specified, see 6.2).

3.7.1.1 For system integrity, heater system shall be able to withstand the following conditions without suffering any damage to the heater system:

(a) Inlet water temperature: 40°F min; 90°F max.
(b) Inlet water pressure: 110 lb/in^2 gauge max.

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(c) Inlet steam temperature: 400°F max.
(d) Inlet steam pressure: 150 lb/in^2 gauge max.
(e) Inlet water pressure fluctuation: 40 lb/in^2 gauge max.

3.7.2 Shock and vibration. Heater system shall meet shock and vibration requirements for grade B deck mounted, class I, of MIL-S-901 and type I of MIL-STD-167-1, respectively (see 4.6.1 and 4.6.2). The tests shall be conducted at normal operating temperatures and pressures. The tested system shall include all components required to make a complete and operable heater system.

3.7.3 Heater inclination. Heater shall operate properly under permanent inclination up to 30 degrees in any direction (see 4.6.3).

3.7.4 Response time. The heater shall be able to generate and deliver continuous and stable temperature hot water at rated temperature and full load after a no load condition within 40 seconds (see 4.6.4).

3.7.5 Intermittent load. Heater shall supply proper stable outlet temperatures when subjected to demands of an intermittent nature, including operation at a fraction of capacity (see 4.6.5).

3.7.6 Partial load. Heater shall meet all the performance requirements specified herein when subjected to partial load conditions (see 4.6.5).

3.7.7 Excess temperature safety. Class A heater system shall demonstrate control of steam and outlet water flow should temperature of the outlet water exceed maximum allowable temperature for all operating conditions within the limits specified in 3.7.1 (see 4.6.6).

3.7.8 Endurance. Heater system shall sustain 100 cycles of heating and cooling (at the maximum rated capacity) without failure to any components or loss of any components (see 4.6.7).

3.7.9 Maintainability. Heater system shall be maintainable within a time equal to MTTR (see 3.4) without personnel hazard (see 4.6.8).

3.8 Hydrostatic testing. Heat exchanger components shall withstand an internal hydrostatic test pressure of 250 lb/in^2 (see 4.6.9).

3.9 Integration requirements. Heater system shall be either pre-assembled or ready for field assembly. Contractor shall identify any assembly requirements.

3.10 Cleaning kits. When specified (see 6.2), a cleaning kit to remove scale buildup due to calcium deposits, shall be furnished.

3.11 Workmanship. The heaters shall be free of visual and operating defects such as sharp edges, dents, scratches, tears in the insulation, paint runs, and malfunctioning controls. Mating surfaces of flanges and mounting pads of supports shall be machined true.

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3.12 Dual temperature systems. When specified (see 6.2), all necessary components for the heater system to service two temperature requirements of 125°F and 160°F (or 180°F) shall be furnished. In such case, all provisions of this specification regarding Class A heaters shall be applied to the portion of the heater system which provides 125°F outlet water temperature.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of the manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.2.1 Instrumentation calibration. Measuring and test equipment used in the performance of tests shall be calibrated in accordance with MIL-STD-45662 (see 6.3). All original "settings," out of tolerance conditions, and retest shall be prepared and certified by the provider and shall be subject to review and approval by the contracting activity.

4.3 First article inspection. Unless otherwise specified (see 6.5), the first heater system of each class and size shall be subjected to first article inspection which shall consist of the examinations and tests specified in table II (see 6.3). Failure of the heater system to pass any of the tests or examinations specified herein shall be cause for rejection.

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TABLE II. First article inspection.

Tests	Requirement	Test method
Examination	3.11	4.5
Hydrostatic Pressure	3.8	4.6.9
Response time	3.7.4	4.6.4
Overtemperature safety device	3.7.7	4.6.6
Endurance	3.7.8	4.6.7
Intermittent load	3.7.5	4.6.5
Partial load	3.7.6	4.6.5
Heater inclination	3.7.3	4.6.3
Shock	3.7.2	4.6.1
Vibration	3.7.2	4.6.2
Maintainability demonstration	3.7.9	4.6.8

4.4 Quality conformance inspection. Quality conformance inspection shall consist of the examinations and tests specified in table III.

TABLE III. Quality conformance inspection.

Tests	Requirement	Test method
Examination	3.11	4.5
Hydrostatic pressure	3.8	4.6.9
Electromagnetic emission and susceptibility	3.6.11.1	4.6.10
Packaging	5.1	4.7

4.4.1 Sampling for quality conformance inspection. A random sample of heaters shall be selected from each lot as specified in table IV. Each heater system shall be examined and measured to verify conformance to the requirements of materials, workmanship and design of this specification, not involving tests. Each heater system in the sample containing one or more defects shall not be offered for delivery (see 6.10).

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TABLE IV. Sampling for quality conformance inspection.

Lot size number of units	Sample size number of units
1 to 6	All
7 to 15	7
16 to 25	10
26 to 40	13
41 to 65	17
66 to 110	22
111 to 180	28
181 to 300	35
300 to 500	45

4.4.2 Lot. For the purposes of quality conformance inspection sampling, a lot is defined as all heaters of the same class, capacity, and construction, using the smae materials and production process, offered for delivery at one time. All units within a lot shall have been fabricated or otherwise assembled within a time span not to exceed one year.

4.5 Examination. Each sample unit selected as specified in 4.4.1 shall be examined for ease of adjustment, fits, leaks, material, finish and conformance to this specification, and as follows:

(a) The external fittings shall be properly secured.
(b) Bolts, nuts, and screws shall be tight. Equipment and accessories shall be properly fastened and secured.
(c) No parts shall be fractured, split, torn, dented, or otherwise damaged to adversely affect serviceability.
(d) There shall be no sharp or ragged edges on any surfaces that may be injurious to personnel.
(e) The temperature control shall be properly set and functioning at the normal operating conditions specified in 3.7.1.

4.5.1 Visual and dimensional examination. Each heater system shall be visually examined to determine conformance with the requirements of table V. Any defects found during inspection as specified in table V shall be cause for rejection.

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TABLE V. Visual and dimensional examination.

Inspection	Requirement
Material	3.2
Temperature adjustments	3.5
Anti-tampering device	3.5.1
Heat exchanger	3.6.2
Shell	3.6.2.1
Head and cover plate(s)	3.6.2.2
Shell liner	3.6.2.3
Heat transfer element	3.6.2.4
End connections	3.6.5
Bosses	3.6.6
Welding and silver brazing	3.6.1
Heater support	3.6.8
Identification and information plates	3.6.9
Steam regulating valve	3.6.10.1
Overtemperature safety shutdowns	3.6.10.2
Safety shutdown (steam side)	3.6.10.2.1
Safety shutdown (water side)	3.6.10.2.2
Pressure relief valve	3.6.10.3
Steam traps and orifices	3.6.10.4
Strainers	3.6.10.5
Temperature and pressure gauges	3.6.10.6
Overtemperature safety control system	3.6.10.7
Gauge panel	3.6.10.8
Electrical equipment	3.6.11
Thermal bulb	3.6.7
Thermal bulb well	3.6.7.2
Integration requirements	3.9
Workmanship	3.11

4.6 Test methods.

4.6.1 Shock test. The shock test shall be performed only after the heater has passed hydrostatic testing and operational tests that indicate the system tested functions properly. The shock test shall be in accordance with MIL-S-901, grade B. The heater system shall be visually inspected for the following conditions occurring during the test or evident at the conclusion of the test:

(a) Any fragmentation or missile effects of parts.
(b) Dislodgement or displacement of any piece, part, or component in such a manner that would be hazardous to personnel in the vicinity of the water heater system or cause an electrical short circuit.
(c) Shearing or tearing of metallic parts that would create projections or sharp edges.
(d) Any external leakage of steam, condensate or potable water that exceeds 140°F in temperature.

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(e) If Class A outlet water exceeds 140°F in temperature.
(f) Any external leakage of potable water at any temperature, when an inspection of internal parts indicates an internal leakage of steam to potable water.

Presence of any material condition specified shall be cause for rejection.

4.6.2 Vibration test. The environmental vibration test shall be conducted in accordance with MIL-STD-167-1 for type I equipment. Acceptance of the water heater system and recirculation system shall be in accordance with the acceptance criteria in MIL-STD-167-1.

4.6.3 Heater inclination. Inclination testing shall be conducted as specified herein. Test shall consist of 3 test phases, where water heater system shall be mounted at 30 degrees inclination about each of 3 axes as shown in figure 4 for durations as specified in table VI. Prior to start of each phase, heater system shall be operated normally for not less than 40 seconds, at which point outlet water shall be rated temperature (see 3.7.1). All normal operating conditions (see 3.7.1) shall be maintained throughout each phase of test. Outlet water flow shall be at rated load. Outlet water shall maintain rated temperature (see 3.7.1) throughout each phase of test. Pressure relief valve shall not relieve and alarm conditions shall not occur during test. Failure of the heater system to meet the specified conditions shall be cause for rejection.

TABLE VI. Heater inclination test.

Phase	Axis of Rotation	Duration (minutes)
1	A	10
2	B	10
3	C	10

4.6.4 Response time testing. Testing response of the heater system to a demand for continuous full load after not less than 60 minutes of no load shall be conducted. For the purposes of this test, no recirculation of outlet water shall be employed. All normal operating conditions (see 3.7.1) shall be maintained throughout the test except as specified herein. Prior to start of test, heater system shall be operated normally for not less than 40 seconds, at which point outlet water shall be rated temperature (see 3.7.1). With outlet water secured, steam at rated pressure and temperature (see 3.7.1) shall be applied at the inlet of steam regulating valve. This condition shall be maintained for not less than 60 minutes, after which the outlet water flow shall be maintained at rated capacity for not less than 5 minutes. Outlet water temperature is permitted to drop below rated during the period of no load. Once water flow is started, outlet water shall achieve and maintain rated temperature (see 3.7.1) after 40 seconds until end of 5 minute period. Pressure relief valve shall not relieve and alarm conditions shall not occur during test. Failure of the heater system to meet the specified conditions shall be cause for rejection.

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4.6.5 Intermittent and partial load testing. Testing of heater system for reliable and safe operation over the wide variety of load conditions experienced shipboard shall be conducted as specified in 4.6.5.1 and 4.6.5.2. For the purposes of these tests, a hot water recirculation system having a recirculating capacity and volume equal to the heater rated capacity (gal/min) shall be employed.

4.6.5.1 Intermittent load testing. Intermittent load testing shall be conducted in one continuous time period, consisting of a series of shorter periods (see table VII) in which specified load conditions apply (see table VII). All normal operating conditions (see 3.7.1) shall be maintained throughout the test. Outlet water shall maintain rated temperature (see 3.7.1) at all times during test. Failure of the heater system to meet the specified conditions shall be cause for rejection.

TABLE VII. Intermittent load test.

Flow rate (percent of rated capacity) (approx.)	Duration (minutes)
0	30
100	30
5	10
10	5
20	5
0	10
40	2
0	60
60	2
0	20
80	2
100	2
100	5

4.6.5.2 Partial load testing. Partial load testing shall be conducted in one continuous time period, consisting of a series of shorter periods not less than 60 seconds duration in which specified load conditions apply. All normal operating conditions (see 3.7.1) shall be maintained throughout the test. The test shall commence with outlet water flow rate corresponding to no load condition. At the beginning of each subsequent 60 second period the outlet water flow shall be increased by 20 percent of rated capacity. The test is complete after final 60 second period elapses (load at 100 percent of rated capacity). Outlet water shall maintain rated temperature (see 3.7.1) at all times during test. Failure of the heater system to meet the specified conditions shall be cause for rejection.

4.6.6 Overtemperature safety device test. The heater system overtemperature safety device test shall be conducted at no load, and at rated load. For the purposes of this test, the thermostatic control shall be defeated. For the purposes of this test, a recirculation system may be employed. The temperature safety device shall be

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set as specified in 3.6.10.7. The test shall consist of increasing the water outlet temperature of each class of heater at a rate of at least 5 degrees per minute. The safety device shall trip and signal the steam regulating valve as specified in 3.6.10.2.1 and 3.6.10.7, and the outlet water safety shutoff valve as specified in 3.6.10.2.2 and 3.6.10.7. The water inlet temperature shall be measured within 12 inches of the water heater inlet. The outlet water temperatures shall be measured not more than 12 inches downstream of the safety devices internal thermal well in the outlet water line. During the test, the water flow rate shall be not less than 50 percent of rated flow of the heater. Inlet and outlet water temperatures shall be taken every 15 seconds during the test. A red telltale light shall illuminate to indicate that a safety shut down has occurred. Failure of the heater system to shut down steam flow and the hot water delivery system as specified herein shall be cause for rejection.

4.6.7 Endurance test. Heater system shall be endurance tested. For the purposes of this test, a recirculation system may be employed. The heater system shall sustain 100 cycles of heating and cooling without failure to any of the components in the heater system. Each cycle shall consist of securing recirculation system (if employed), securing steam, allowing admission of a quantity of inlet water at 40°F at rated flow for 1 minute, then reactivate steam and recirculation system (if employed), allowing heater to operate normally. This operation shall continue until the desired delivery water temperature is obtained and maintained for not less than 5 minutes. Every tenth cycle the heater system shall be shut down, drained, and flushed with cool water to lower the temperature of the unit to room temperature, to complete the cycle. After completion of the endurance test, the system shall be hydrostatically tested (see 4.6.9). Each heater system component including all valves shall be disassembled and examined for deterioration of the interior surfaces and exposed elements. Any visible sign of wear shall be cause for rejection. Failure of water heater systems to meet all requirements shall be cause for rejection (see 3.4).

4.6.8 Maintainability demonstration. The capability to maintain, disassemble, and repair the heater system shall be demonstrated to a Government representative (see 6.3). The heat exchanger and valves (see 4.6.8.1) shall be disassembled, reassembled, and replaced. The controls and instruments shall be removed and replaced under the observation of the Government representative. The demonstration shall show that access to and replacement of components is possible without personnel hazard. The personnel and specific types of tools used in the demonstration shall be identified. Failure of the system to be disassembled and reassembled without personnel hazard in accordance with procedures outlined in the technical manual shall be cause for rejection. In addition, the demonstration shall verify that the MTTR requirement specified in 3.4 is satisfied by the design and construction of the heater assembly.

4.6.8.1 Disassembly and reassembly procedure. The degree of disassembly and reassembly procedure shall include the following items:

(a) Heat exchanger.
(b) Regulating valves.
(c) Solenoid valves.
(d) Relief valve.
(e) Steam traps.

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(f) Overtemperature sensor.
(g) Strainers.

4.6.9 Hydrostatic testing. Heat exchanger components including shells, heads, cover plates, and end connections shall be hydrostatically tested to 250 lb/in^2 gauge on each side (water and steam) separately to verify system integrity (see 3.8). Full test pressure shall be applied to the water side with the steam side drained of water and vice versa. The pressure shall be isolated and maintained in these units for not less than 30 minutes. The coil side of the heat exchanger may be tested before it is assembled within shell, but the heat exchanger shall be fully assembled prior to hydrostatically testing the shell side. The heater system's temperature and pressure gauges shall be isolated during the test. The failure to maintain 250 lb/in^2 gauge in a unit for 30 minutes resulting from visible evidence of leakage, permanent distortion, or collapse shall be cause to reject the unit.

4.6.10 Electromagnetic emission and susceptibility test. The electromagnetic test shall be in accordance with MIL-STD-1399 (see 6.3) and requirements CE101, CE102, CS104, RE101, RE102, RS101, and RE103 of MIL-STD-461.

4.7 Inspection of packaging. Sample packages and packs, and the inspection of the preservation, packing, and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition. For the extent of applicability of the packaging requirements of referenced documents listed in section 2, see 6.9.)

5.1 Packaging requirements. Heaters, accessories, and technical data, shall be preserved level A, C, or commercial, packed level A, B, C, or commercial as specified (see 6.2), marked in accordance with MIL-E-17555, and shall include bar codes and applicable packaging acquisition options therein as specified (see 6.2). In addition, for Navy acquisitions, the following applies:

(a) Navy shipboard stowage fire-retardant requirements.

(1) Treated lumber and plywood. Unless otherwise specified (see 6.2), all lumber and plywood including laminated veneer material used in shipping containers and pallet construction, members, blocking, bracing, and reinforcing shall be fire-retardant treated material conforming to MIL-L-19140 as follows:

Levels A and B - Type II - weather resistant.
Category 1 - general use.

Level C - Type I - non-weather resistant.
Category 1 - general use.

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(2) Fiberboard. Unless otherwise specified (see 6.2), fiberboard used in the construction of class-domestic, non-weather resistant fiberboard and cleated fiberboard boxes including interior packaging forms shall meet the flame spread and the specific optic density requirements of PPP-F-320 and amendments thereto.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Non-storage type, quick recovery, steam water heater systems are to be used on board ships where continuous hot water generation is required, space is at a minimum, low weight is required, and low steam consumption is desirable.

6.2 Acquisition requirements. Acquisition documents must specify the following:

(a) Title, number, and date of this specification.
(b) Class and rated capacity of heater (see 1.2 and 1.2.1).
(c) Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
(d) When first article inspection is required (see 3.1).
(e) When class A heaters require an anti-tampering device (see 3.5.1).
(f) When end connections are other than as specified (see 3.6.5).
(g) Electrical equipment if other than as specified (see 3.6.11).
(h) When hazard analysis is required (see 3.6.13.3).
(i) Not used.
(j) When heater performance condition parameters are other than as specified (see 3.7.1).
(k) Inlet water temperature if other than as specified (see 3.7.1(b)).
(l) Inlet water pressure, if other than as specified (see 3.7.1(c)).
(m) Outlet water temperature if other than as specified (see 3.7.1(h)).
(n) Inlet steam temperature if other than as specified (see 3.7.1(e)).
(o) Not used.
(p) Inlet steam pressure if other than as specified (see 3.6.3 and 3.7.1(f)).
(q) When cleaning kits are required (see 3.10).
(r) Levels of preservation and packing required (see 5.1).
(s) Marking, bar codes, and options required (see 5.1).
(t) Navy shipboard stowage fire-retardant requirements if other than as specified (see 5.1).
(u) When two temperature systems are required (see 3.12).

6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DID's) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DID's are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements

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List (DD Form 1423) must be prepared to obtain the data, except where DoD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference Paragraph	DID Number	DID Title	Suggested Tailoring
3.3.1 and appendix	DI-DRPR-80651	Engineering drawings	----
3.4	DI-RELI-80250	Reliability test plan	----
3.4	DI-RELI-80251	Reliability test procedures	----
3.4	DI-RELI-80252	Reliability test reports	----
3.6.13.3	DI-SAFT-80101	System safety hazard analysis report	Task 202
4.2.1 and 4.3	DI-MISC-80678	Certification/data report	10.3.1 does not apply
4.4	DI-NDTI-80809	Test/inspection reports	----
4.6.8	DI-R-5318	Maintainability demonstration plan	----
4.6.8	DI-R-5192	Maintainability demonstration report	----
4.6.10	DI-EMCS-80200	Electromagnetic interference test report	----
6.4.1	DI-H-5521	Preparing technical presentations for video taping	----

The above DID's were those cleared as of the date of this specification. The current issue of DoD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DID's are cited on the DD Form 1423.

6.4 Technical manuals. The requirement for technical manuals should be considered when this specification is applied on a contract. If technical manuals are required, military specifications and standards that have been cleared and listed in DoD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL) must be listed on a separate Contract Data Requirements List (DD Form 1423), which is included as an exhibit to the contract. The technical manuals must be acquired under separate contract line item in the contract.

6.4.1 Audiovisual tape. When required, an audiovisual training tape should be prepared (see 6.3). The training tape should be a one-inch video master which illustrates the operation and function, and demonstrates maintenance and repair of the units. The master tape should illustrate the operation, function, maintenance, and repair of class A, class B, and class C heaters. The tape must be acquired under a separate contract line item in the contract.

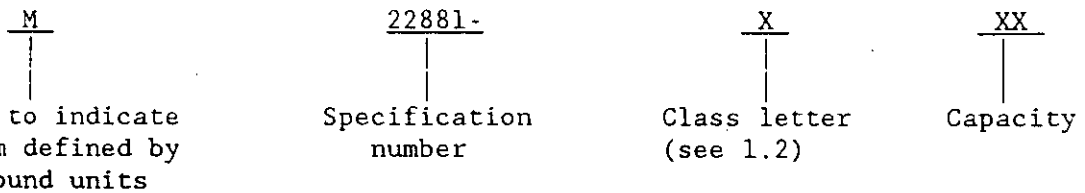
6.5 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a pre-production sample, a first article sample, a first production item, a sample selected from the first production items, or a standard production item from the contractor's current inventory (see 3.1), and the number of items to be tested as

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specified in 4.3. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.6 Sub-contracted material and parts. The packaging requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.7 Part or Identifying Number (PIN). The PINs to be used for heaters acquired to this specification are created as follows:



An abbreviated class, capacity designation consisting of the class letter and the capacity (example: A10) may be used.

6.8 NAVSEA approval and direction. Deviations from specified materials, procedures, and requirements and selection of specific alternative materials and procedures require NAVSEA approval or direction. Requests should include supporting documentation.

6.9 Provisioning. Provisioning Technical Documentation (PTD), spare parts, and repair parts should be furnished as specified in the contract.

6.9.1 When ordering spare parts or repair parts for the equipment covered by this specification, the contract should state that such spare parts and repair parts should meet the same requirements and quality assurance provisions as the parts used in the manufacture of the equipment. Packaging for such parts should also be specified.

6.10 Lot acceptance and rejection criteria. The establishment of an Acceptable Quality Level (AQL) should be considered when this specification is applied on a contract. The AQL should be based on suggested lot acceptance and rejection criteria shown in table VIII. If the number of defective systems in any sample exceeds acceptance criteria for that sample, the lot represented by the sample should not be offered for delivery.

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TABLE VIII. Lot acceptance and rejection criteria.

Lot size number of units	Sample size number of units	Defective units	
		Acceptance number	Rejection number
1 to 6	All	---	---
7 to 15	7	0	1
16 to 25	10	0	1
26 to 40	13	0	1
41 to 65	17	1	2
66 to 110	22	1	2
111 to 180	28	2	3
181 to 300	35	2	3
300 to 500	45	3	4

6.11 Subject term (key word) listing.

Heat exchangers
Non-personnel use heaters
Personnel use heaters

6.12 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Preparing activity:
Navy - SH
(Project 4520-N002)

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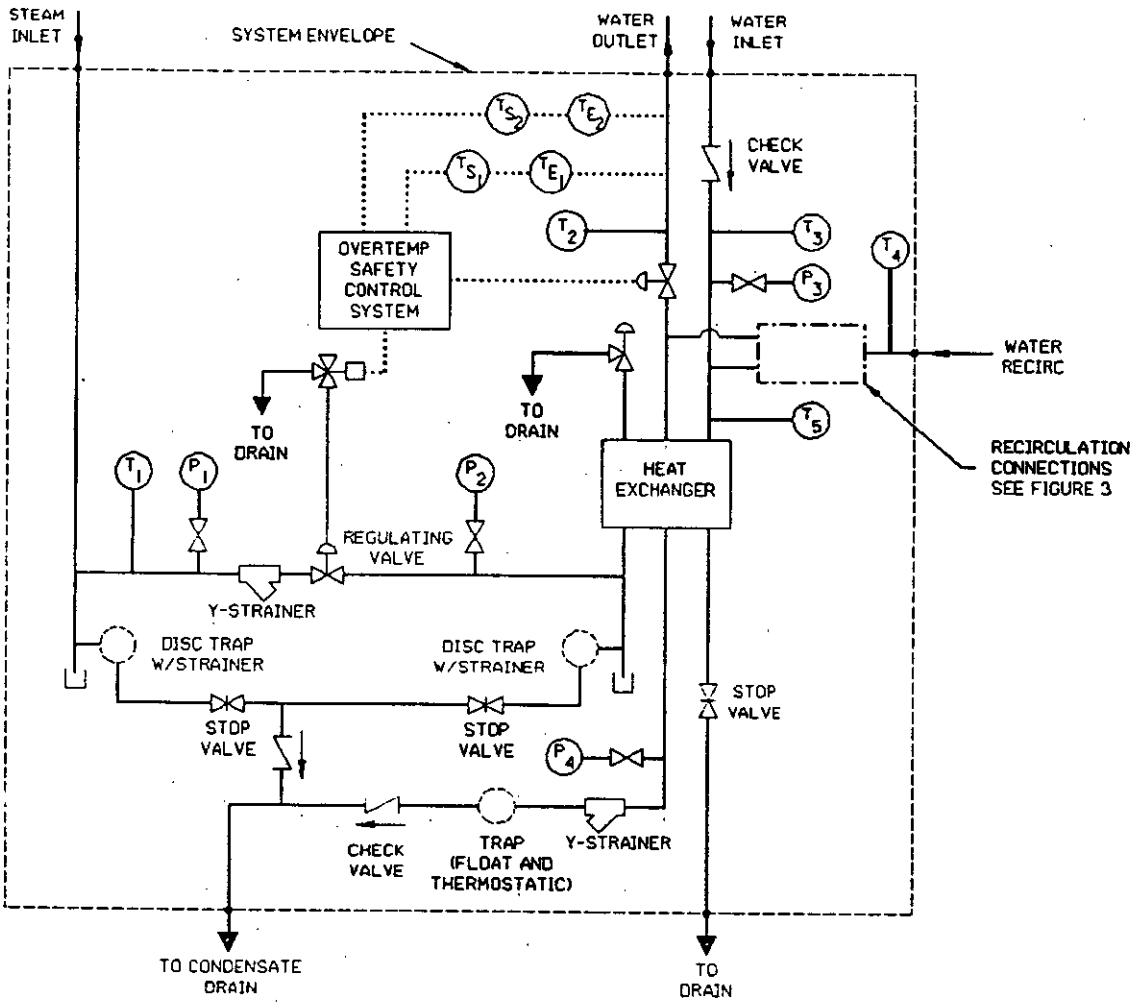


Figure 1. Typical class A heater system.

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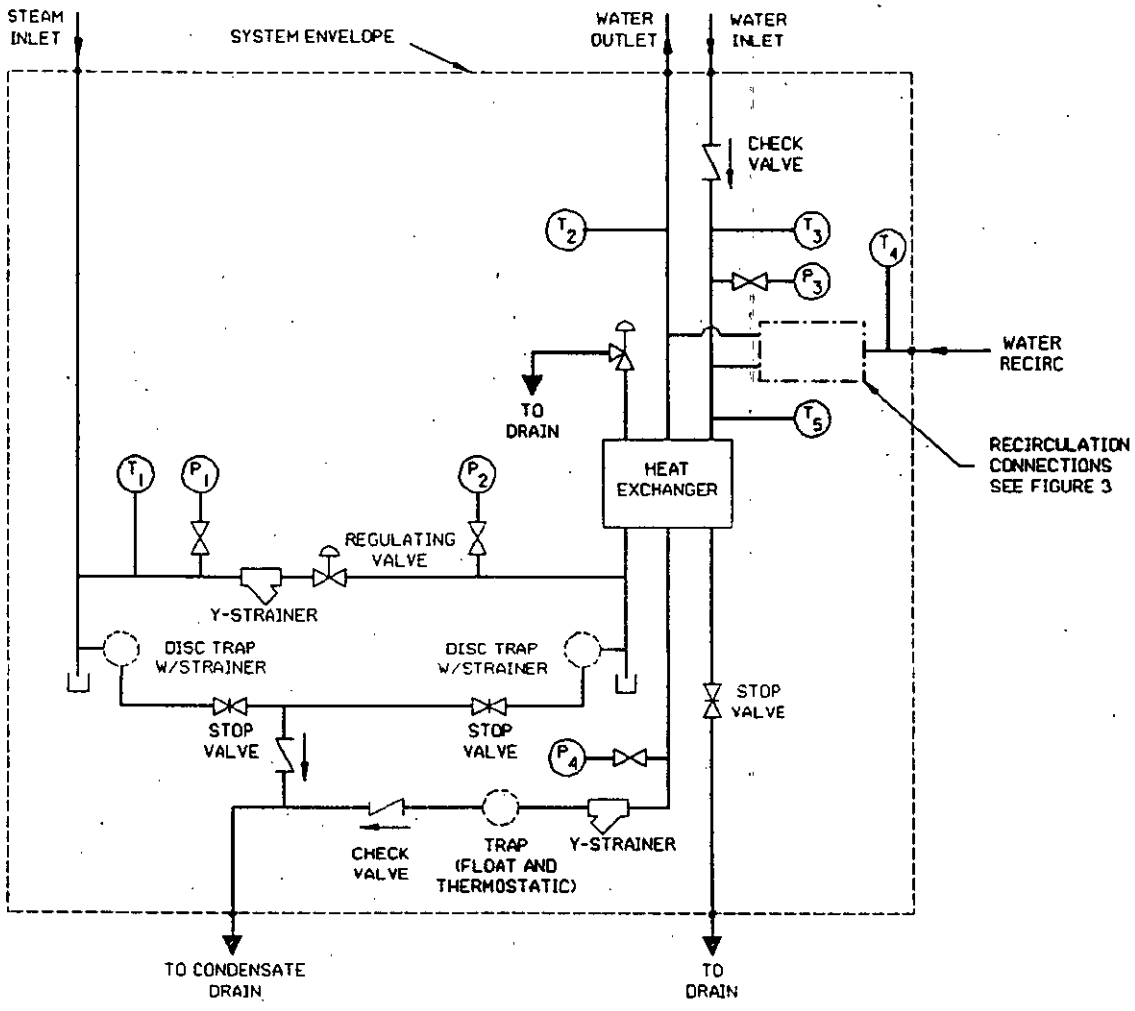


Figure 2. Typical class B or C heater system.

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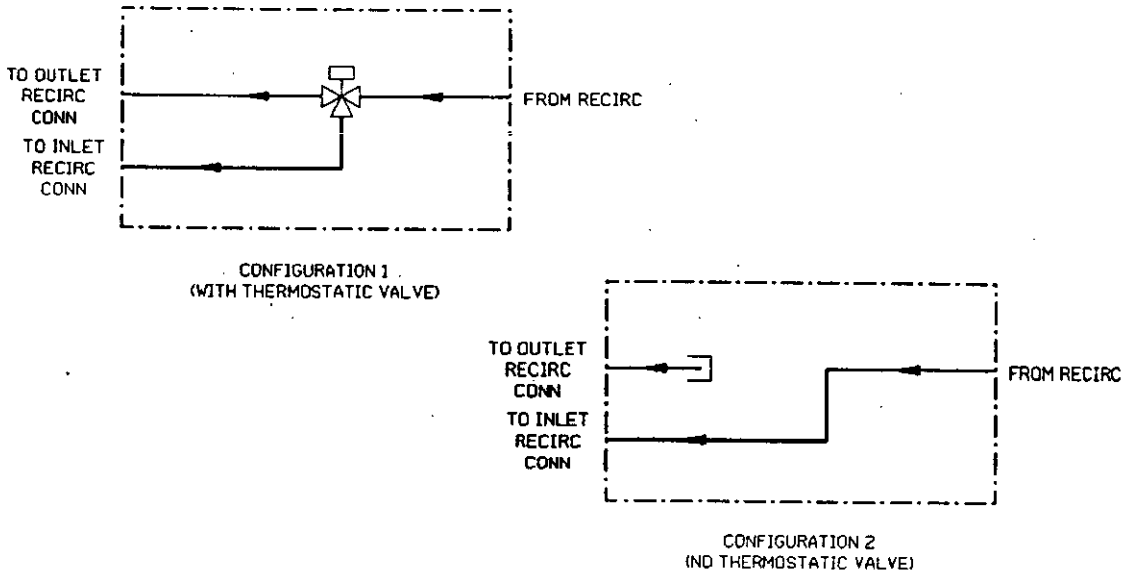


Figure 3. Recirculation connections for typical heater system.

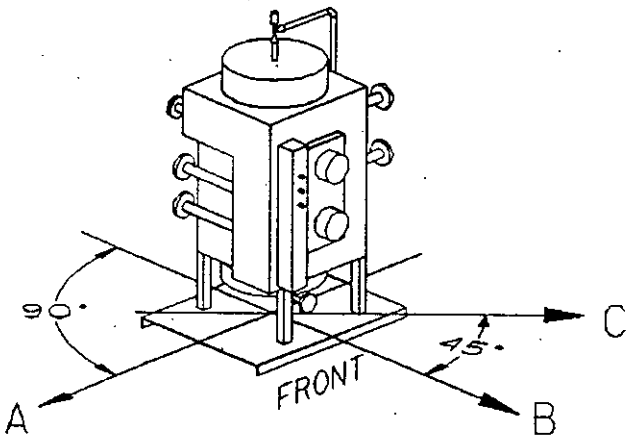


Figure 4. Heater inclination axes for testing.

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APPENDIX

ENGINEERING DRAWINGS TECHNICAL CONTENT REQUIREMENTS

10. SCOPE

10.1 Scope. This appendix covers information that shall be included on the drawings when specified in the contract or order. This appendix is mandatory only when data item description DI-DRPR-80651 is cited on the DD Form 1423.

20. APPLICABLE DOCUMENTS

This section is not applicable to this appendix.

30. DRAWING CONTENTS

30.1 Drawings. Heater and heater component drawings shall be levels 2 or 3, as applicable.

30.1.1 Level 2 drawings. Level 2 heater and heater component drawings shall be sufficient to permit evaluation, and so forth.

30.1.2 Level 3 drawings. Level 3 drawings shall include detail manufacturing parts, subassembly and final assembly drawings, and manufacturing and assembly procedures.

30.2 Limited rights legend. When the government has only limited rights in the data shown on the drawings, as determined by the contractual provisions regarding rights in technical data, the drawings furnished may be marked with the following restrictive legend:

"Furnished under United States Government Contract No._____.
Shall not be either released outside the Government, or used, duplicated, or disclosed in whole or in part for manufacture or procurement, without the written permission of _____, except for: (a) emergency repair or overhaul work by or for the Government, where the item or process concerned is not otherwise reasonably available to enable timely performance of the work; or (b) release to a foreign government, as the interests of the United States may require, provided that in either case the release, use, duplication, or disclosure hereof shall be subject to the foregoing limitations. The legend shall be marked on any reproduction hereof in whole or in part."

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-H-22881E(SH)	1995 MAY 05

3. DOCUMENT TITLE

HEATER SYSTEMS, WATER, STEAM HEATED NON-STORAGE TYPE

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
Technical Point of Contact (TPOC) Mr. Charlie Brady, PSNS 260.42	(1) Commercial (206) 476-1487	(2) AUTOVON 439-1487
c. ADDRESS (Include Zip Code) Commander, Puget Sound Naval Shipyard ATTN: PSNS 260.42, 1400 Farragut Ave. Bremerton, WA 98314-5001	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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